CHAPTER 18

ADC & DAC

Analog ↔ digital conversion · resolution · quantization · sampling · DAC types · flash, counter, SAR ADC

QUICK TRICK

Lines to learn first: n bits give 2^n **levels**. **Step (1 LSB) =** $V_{ref} / 2^n$. **Flash ADC uses** $2^n - 1$ **comparators**. **SAR ADC takes n clock pulses. Sample at least twice the highest frequency.**

18.1 Why do we need ADC and DAC?

● **LEVEL 1 · BEGINNER**

The real world is **analog** (sound, temperature, light, voltage). Computers are **digital** (0 and 1). We need a translator in both directions:

Converter	Full name	Direction	Example
ADC	Analog-to-Digital Converter	Real world → computer	Microphone → digital audio
DAC	Digital-to-Analog Converter	Computer → real world	Digital audio → speaker

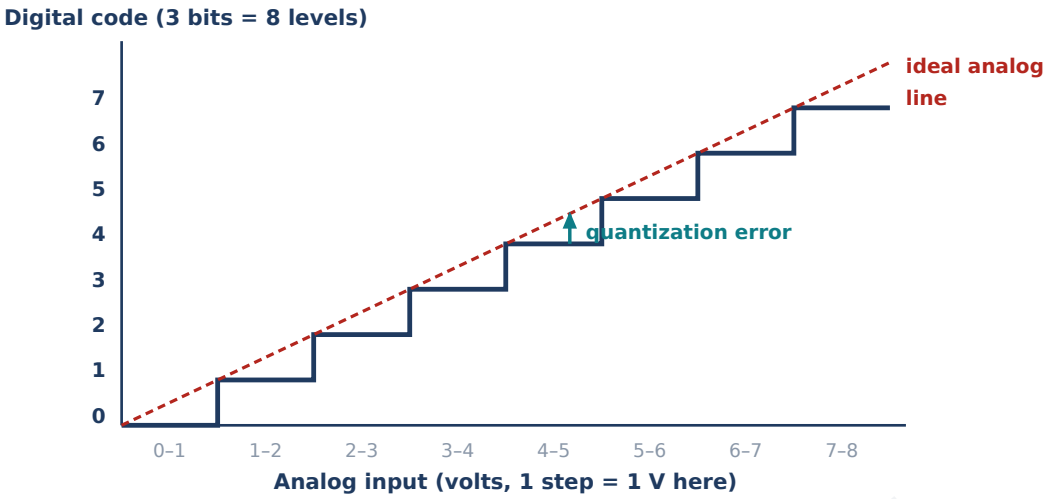
```
Sensor -> [ ADC ] -> Digital system (computer) -> [ DAC ] -> Speaker / motor
```

Analogy. Measuring a person's height with a ruler marked only in whole centimetres. You must round to the nearest mark. More marks (more bits) mean a more exact answer.

18.2 Important terms

● **LEVEL 2 · EXAM**

Term	Simple meaning	Formula
Resolution (levels)	How many different digital values the converter can use	2^n levels for n bits
Step size (1 LSB)	The smallest change in the analog value that changes the code	$V_{ref} / 2^n$ (full-scale / 2^n)
Quantization	Rounding the analog value to the nearest allowed level	—
Quantization error	The difference between the real value and the rounded value	Maximum ±½ LSB (when rounding to nearest)
Sampling	Measuring the signal at regular times	Sampling rate f_s
Nyquist rule	To rebuild the signal, sample at least twice the highest frequency	$f_s \geq 2\, f_{max}$
Conversion time	Time the ADC needs to give one result	Max sample rate = 1 / conversion time



The black staircase is the digital output. The red dashed line is the real analog input. The **gap between them** is the quantization error. It is at most one step (or half a step when we round to the nearest level).

Example 1 (Level 2): step size of an 8-bit ADC, full-scale 5 V

Given: n = 8, V_{ref} = 5 V. **Concept:** step = $V_{ref} / 2^n$.

Calculation: 5 / 256 = 0.01953 V = **19.53 mV**. Maximum quantization error = ½ step = **±9.77 mV**.

Example 2 (Level 3): how many bits do I need?

Given: input range 0 to 10 V, required resolution 10 mV or better.

Concept: number of steps = 10 V / 10 mV = 1000. We need $2^n \geq 1000$.

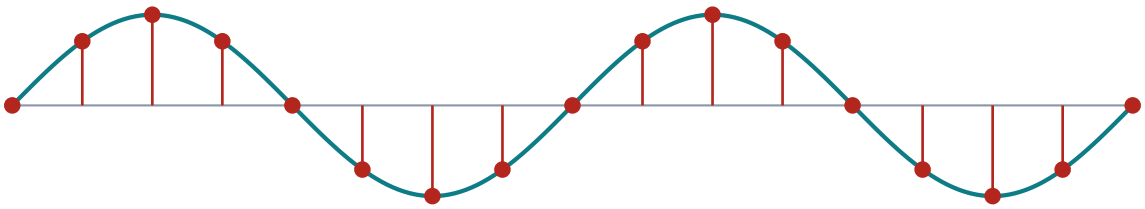
Calculation: $2^9 = 512$ (too small), $2^{10} = 1024$ (enough).

Answer: 10 bits.

QUICK TRICK

Each extra bit doubles the number of levels and halves the step size. It also improves the signal-to-noise ratio by about **6 dB** (SNR ≈ 6.02n + 1.76 dB: 8 bits ≈ 49.9 dB, 16 bits ≈ 98 dB).

Sampling and the Nyquist rule



Sampling: measure the signal at regular times (the dots). Here we take 8 samples per cycle.

If we sample too slowly, different signals look the same (this is called **aliasing**) and the signal cannot be rebuilt. To avoid it, $\mathbf{f_s \geq 2\, f_{max}}$. Example: audio up to 20 kHz needs at least 40 kHz (CDs use 44.1 kHz). A signal with highest frequency 5 kHz needs at least **10 kHz**.

COMMON MISTAKE

Confusing the sampling rate with the signal frequency. The rule is **twice the highest frequency** in the signal, not the signal frequency itself.

18.3 DAC: digital to analog

● LEVEL 2 · EXAM

Idea. Each bit of the digital word adds a **weighted** amount to the output. The MSB adds the most, the LSB the least.

```
V_out = D x (V_ref / 2^n)            D = decimal value of the digital input
For 4 bits:  V_out = V_ref ( b3/2 + b2/4 + b1/8 + b0/16 )
```

Example 3 (Level 2): 4-bit DAC, V_{ref} = 5 V, input 1010

Given: D = 1010 = 10, n = 4. **Concept:** $V_{out} = D \times V_{ref} / 2^n$.

Calculation: 10 × 5 / 16 = **3.125 V**. (One step = 5/16 = 0.3125 V.)

Example 4 (Level 3): 8-bit DAC, V_{ref} = 10 V

Step = 10 / 256 = 39.06 mV.

Input 10000000 (128): V_{out} = 128 × 39.06 mV = **5 V** (the MSB alone gives half of V_{ref}).

Input 00000001: V_{out} = **39.06 mV** (one step).

QUICK TRICK

MSB gives $V_{ref}/2$, the next bit $V_{ref}/4$, then $V_{ref}/8$… Just add the weights of the bits that are 1. Full-scale output = $V_{ref}(1 - 2^{-n})$, one step below V_{ref}.

Binary-weighted resistor DAC

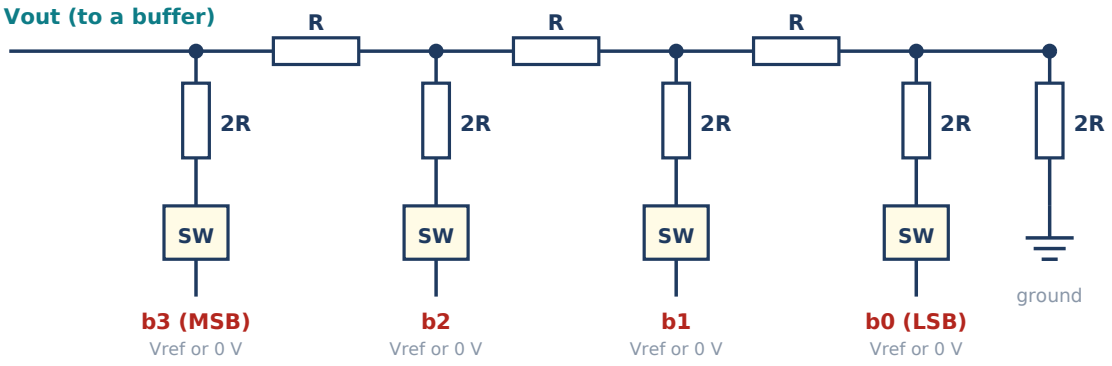
Each bit switches a resistor into an op-amp summing circuit. The resistors are **R, 2R, 4R, 8R…** (MSB has the smallest resistor, so it adds the most current). With feedback resistor $R_f = R/2$: **$V_{out} = -V_{ref}$ (b3/2 + b2/4 + b1/8 + b0/16)** (the op-amp inverts the sign).

COMMON MISTAKE

Problem: for n bits you need resistors from R up to 2^{n-1}R. An 8-bit DAC needs a ratio of 128 : 1, and every resistor must be very accurate. This is hard to build in a chip.

R-2R ladder DAC

Idea. Use only **two** resistor values, R and 2R, in a ladder. Each bit switch connects its 2R leg to V_{ref} (bit = 1) or to ground (bit = 0). At every node the ladder divides the voltage by 2, which creates the binary weights.



	Binary-weighted DAC	R-2R ladder DAC
Resistor values	n different values (R to 2^{n-1}R)	**Only 2** values (R and 2R)
Accuracy needed	Very high for the large resistors	Easy to match
Building in an IC	Difficult for many bits	**Easy** (the usual choice)
Resistors for n bits	n (plus R_f)	**2n** (n − 1 of R and n + 1 of 2R)

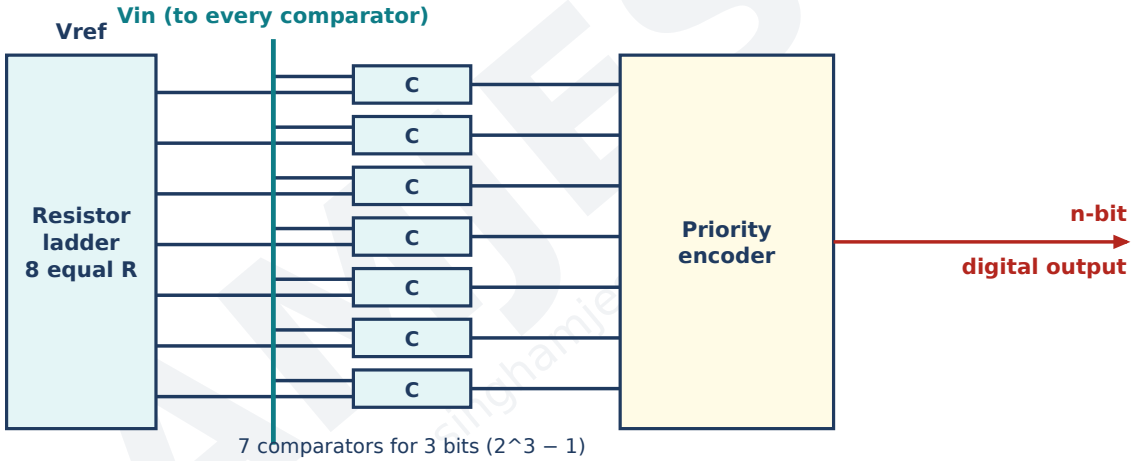
QUICK TRICK
R-2R counting: an n-bit ladder has **2n** resistors. 8 bits → 16 resistors (7 of R and 9 of 2R). The last node has two 2R: one for the LSB switch and one to ground (the termination).

18.4 ADC: analog to digital

● **LEVEL 2 · EXAM**

Flash (parallel) ADC — the fastest

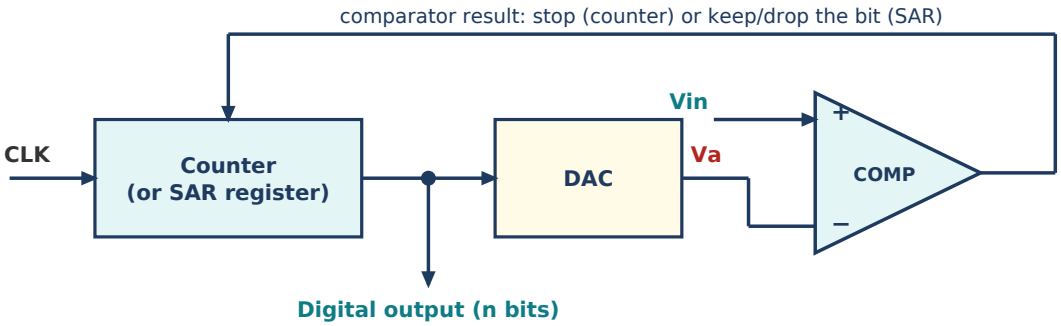
Idea. Compare the input with **every** possible level at the same time. A resistor ladder makes the $2^n - 1$ reference voltages. **$2^n - 1$ comparators** check them all. A priority encoder converts the result into a binary number.



QUICK TRICK
Flash ADC: $2^n - 1$ comparators, 1 clock cycle, fastest, but expensive. 8 bits → **255** comparators. 10 bits → 1023. The number of comparators doubles with every extra bit, so flash ADCs are used for only a few bits (video, radar).

Counter-type (digital ramp) ADC and successive-approximation (SAR) ADC

Both use a **DAC and a comparator in a feedback loop**. The control block guesses a digital number, the DAC turns it into a voltage V_a, and the comparator says if V_a is below or above the input V_{in}.



	Counter-type ADC	SAR ADC
Control block	A binary counter	A successive-approximation register
Method	Count up from 0 until $V_a \geq V_{in}$	**Binary search**: decide one bit at a time, MSB first
Conversion time	**Up to 2^n** clock pulses (average 2^{n-1}). Not constant.	**n** clock pulses (always)
Speed	Slow	Fast

QUICK TRICK

SAR rule: for each bit from the MSB down: **set it to 1, let the DAC make V_a. If $V_a \leq V_{in}$ keep the 1, otherwise change it back to 0.** It always finishes in **n clock pulses**, because every bit is decided once.

Example 5 (Level 3): SAR ADC, 4 bits, V_{ref} = 16 V (step = 1 V), V_{in} = 11.3 V

Each trial sets one more bit to 1 and checks the DAC voltage V_a:

Bit tried	Trial code	DAC output Va	Va ≤ 11.3 V ?	Code kept so far
bit 3	1000	8 V	Yes → keep 1	1000
bit 2	1100	12 V	No → back to 0	1000
bit 1	1010	10 V	Yes → keep 1	1010
bit 0	1011	11 V	Yes → keep 1	1011

Result: **1011 = 11 V** (the real value was 11.3 V; the missing 0.3 V is the quantization error). It took **4 clock pulses**.

Example 6: conversion times

SAR, 8 bits, 2 MHz clock: 8 pulses × 0.5 µs = **4 µs**.

Counter-type, 8 bits, 1 MHz clock: worst case 2^8 = 256 pulses = **256 µs**, average 128 µs.

Flash: **1 clock pulse**.

Dual-slope ADC (integrating)

It charges a capacitor with the input for a fixed time, then discharges it with a reference and counts the time. It is **very accurate and rejects noise** but **slow**. Used in digital voltmeters.

ADC type	Speed	Hardware	Conversion time	Used for
Flash	Fastest	$2^n - 1$ comparators (large)	1 clock	Video, radar
SAR	Fast	1 comparator + DAC + register	n clocks	General purpose, data acquisition
Counter-type	Slow	1 comparator + DAC + counter	Up to 2^n clocks	Simple, low-speed
Dual-slope	Slowest	Integrator, counter	Very long	Digital voltmeters, accurate

EASY MEMORY

Speed order: Flash > SAR > Counter > Dual-slope. **Accuracy and noise rejection** is best in dual-slope. **Cost and size** is highest for flash.

Quick-Solve Tricks — Summary

Topic	Trick
Levels and step	2^n levels. Step = V_ref / 2^n.
DAC output	V_out = D × step. Add the weights of the 1 bits (V_ref/2, /4, /8…).
Quantization error	At most ½ step (rounding to nearest).
Bits needed	Number of steps = range / resolution. Take the next power of 2.
Nyquist	$f_s \geq 2 f_{max}$.
Flash	$2^n - 1$ comparators, 1 clock.
SAR	n clocks. Try each bit from the MSB: keep it if $V_a \leq V_{in}$.
Counter-type	Up to 2^n clocks, average 2^{n-1}.
R-2R	Only R and 2R. 2n resistors for n bits.
Max sample rate	1 / conversion time.

GATE QUICK CHECK — Chapter 18

```
Levels = 2^n          Step = Vref / 2^n          Max quantization error = step / 2
DAC : Vout = D x Vref / 2^n                      Nyquist : fs >= 2 fmax
Flash : 2^n - 1 comparators     SAR : n clocks     Counter ADC : up to 2^n clocks
```

Traps: 2^n vs $2^n - 1$ (levels vs comparators) · counter-type time is not constant · R-2R uses only two resistor values · the Nyquist rate is twice the **highest** frequency.

Practice questions

● LEVEL 1 · BEGINNER

Q1. How many levels does an 8-bit ADC have?

Q2. How many comparators does an 8-bit flash ADC need?

Q3. Which ADC type is the fastest?

● LEVEL 2 · EXAM

Q4. Find the step size of a 10-bit ADC with a 5 V full-scale range.

Q5. A 4-bit DAC has V_{ref} = 5 V. Find V_{out} for the input 1010.

Q6. A signal has a highest frequency of 5 kHz. What is the minimum sampling rate?

● LEVEL 3 · GATE

Q7. An 8-bit SAR ADC uses a 2 MHz clock. How long does one conversion take?

Q8. An 8-bit counter-type ADC uses a 1 MHz clock. Find the worst-case and the average conversion time.

Q9. How many bits are needed to measure 0–10 V with a resolution of 10 mV or better?

Q10. A 12-bit ADC has a full-scale range of 4.096 V. Find the step size and the maximum quantization error.

Q11. A 4-bit SAR ADC has V_{ref} = 16 V. The input is 11.3 V. Find the output code.

Q12. How many resistors are in an 8-bit R-2R ladder DAC? An 8-bit DAC has V_{ref} = 10 V. What is the output for the input 10000000?

Answers

Q	Solution
1	2^8 = **256** levels.
2	$2^8 - 1$ = **255** comparators.
3	**Flash** (parallel) ADC.
4	5 V / 2^{10} = 5 / 1024 = **4.88 mV**.
5	10 × 5 / 16 = **3.125 V**.
6	$f_s \geq 2 \times 5$ kHz = **10 kHz**.
7	n = 8 clock pulses. Period = 1/2 MHz = 0.5 µs. Time = 8 × 0.5 µs = **4 µs**.
8	Worst case = 2^8 = 256 pulses = 256 × 1 µs = **256 µs**. Average = half = **128 µs**.
9	Steps = 10 V / 10 mV = 1000. 2^9 = 512 < 1000 ≤ 2^{10} = 1024 → **10 bits**.
10	Step = 4.096 / 2^{12} = 4.096 / 4096 = **1 mV**. Maximum error = ½ step = **0.5 mV**.
11	Step = 1 V. Trials: 8 V (keep), 12 V (drop), 10 V (keep), 11 V (keep) → **1011** (= 11 V).
12	2n = **16** resistors (7 of R, 9 of 2R). Step = 10/256; input 128 → 128 × 10/256 = **5 V** (half of V_{ref}).

Next: Chapter 19 — Logic Families (TTL, ECL, MOS, CMOS; propagation delay, power, fan-out, noise margin, comparison).